

ODD 2024 Tutorial Sheets – 3

Software Development Fundamentals – I (15B11CI111)

COURSE OUTCOMES		COGNITIVE LEVELS
CO1	Explain the logic for solving problems considering various phases of software development life cycle and depicting them using algorithms and flowcharts	Understand (Level 2)
CO2	Explain basics of C programming concepts to make decision for solving problems	Understand (Level 2)
CO3	Demonstrate and contrast different methods for writing modular programs in C	Understand (Level 2)
CO4	Use various C programming constructs to implement iteration, and recursion	Apply (Level 3)
CO5	Apply and implement arrays, pointers, structures and file handling for solving real-world problems	Apply (Level 3)

Note: Students are advised to submit their solutions to their respective tutorial faculties

Before attempting the following questions, discuss the doubts in earlier tutorials (Tutorial 1 and Tutorial 2), if any.

Q1. [CO2] What will be the output of following program?

```
#include<stdio.h>
int main()
{
    printf("%d\t",sizeof(2.5));
    printf("%d\t",sizeof(2));
    printf("%d\t%d",sizeof('A'),'A');
    return 0;
}
```

Q2. [CO2]What will be the output of the following program?

```
#include<stdio.h>
int main(){
    float me = 5.25;
    double you = 5.25;
    if(me == you)
        printf("Hey");
    else
        printf("Bye");
    return 0;
}
```

Q3. [CO2]What will be the output of the following program?

```
#include <stdio.h>
int main()
{
    int a=11, b=22, c;

    c = a + b + a++ + b++ + ++a + ++b;
    printf("a=%d",a);
    printf("b=%d",b);
    printf("c=%d",c);

}
```

Q4. [CO2] Show the output of the following code:

```
#include <stdio.h>
int main() {
    char ch = 'B';
    printf("%c\n", ch);
    int x = 45, y = 90;
    printf("%d\n", x);
    printf("%i\n", y);
    float f = 12.67;
    printf("%f\n", f);
    printf("%e\n", f);
    int a = 67;
    printf("%o\n", a);
    printf("%x\n", a);
    char str[] = "Hello World";
    printf("%s\n", str);
    printf("%20s\n", str);
    printf("%-20s\n", str);
    printf("%20.5s\n", str);
    printf("%-20.5s\n", str);
    return 0;
}
```

Q5. [CO2]Which one of the following operator performs operations with only integer operands?

- A. +
- B. *
- C. /
- D. %

Q6. [CO2] Which one of the following statements are true in context of || (logical OR) operator

- S1: Evaluation of the expression involving || operators only will stop if one of its components is evaluated as true
- S2: Evaluation of the expression involving || operators only will stop if one of its components is evaluated as false
- S3: Evaluation of the expression involving || operators only takes place from right to left
- S4: Evaluation of the expression involving || operators only takes place from left to right

- A. S1 & S2
- B. S1 & S3
- C. S1 & S4
- D. None of the listed options

Q7. [CO2] Which one of the following is the output of the program given below?

```
#include<stdio.h>
void main()
{
int A = - -2
printf(“%d”, A);
}
```

- A. -2
- B. 2
- C. Will give error
- D. None of the listed options

Q8. [CO2] Which one of the following is the output of the program given below?

```
#include <stdio.h>
int main()
{
int x, y, z;
x = 20;
y = 30;
z = 10;

if(x > y, x > z)
printf("IF BLOCK");
else
printf("ELSE BLOCK");
return 0;
```

```
}
```

- A. IF BLOCK
- B. ELSE BLOCK
- C. Will give error
- D. None of the listed options

Q9. [CO2] Which one of the following is the output of the program given below?

```
#include <stdio.h>
int main()
{
    int w, x, y, z;
    x = 20;
    y = 30;
    z = 10;
    w = (x = z, y += x, z = x + y + z);
    printf("%d %d %d %d", w, x, y, z);
}
```

- A. 60, 10, 40, 60
- B. 10, 10, 40, 60
- C. 10, 10, 30, 60
- D. None of the listed options

Q10. [CO2] Which one of the following is not a valid declaration of a variable in C?

D1: signed int x;
D2: unsigned short y;
D3: unsigned short int z;
D4: signed short w;

- A. D2
- B. Both D2 and D4
- C. All declarations are valid
- D. None of the listed options

Q11. [CO2] What will be output of the following program?

```
int main()
{
    float w, x;
    double y, z;
    w = 20 / 3;
    x = 20.0 / 3;
    y = 20 / 3;
    z = 20.0 / 3;
```

```
if(w == y)
printf("Same");
else
printf("Different");
if(x == z)
printf("\t Same");
else
printf("\t Different");
if(w == x)
printf("\t Same");
else
printf("\t Different");
return 0;
}
```

- A. Same SameSame
- B. Same DifferentDifferent
- C. Different SameSame
- D. Different DifferentDifferent